

PRIM RATTANAK

Robotic and Automation Engineering

FRESH GRADUATE

ABOUT ME

Robotics and Automation Engineering graduate, currently part of the DCLab UAV team. Passionate about advancing UAV technology, eager to apply skills in a professional setting, collaborate with experts, and innovate in automation and robotics.

EXPERIENCE

DYNAMICS AND CONTROL LABORATORY

Institute of Technology of Cambodia | 2022-2025

- **Design and develop a 7-DOF robotic manipulator** as the main thesis project, covering kinematics, motion planning, and control.
- **Internship in Camflytec:** Development of a Quadruped Robot with Embedded Thermal Imaging and Hardware Programming for Advanced Applications
- **Develop and refine a VTOL (Vertical Take-Off and Landing)** project for improved aerial mobility.
- **Apply AI and machine learning techniques** to enhance autonomous drone operations using Jetson Nano and Raspberry Pi.
- **Implement movement detection**, object detection, and auto-landing algorithms for drones.
- **Experiments with OpenCV** for face detection, face tracking, and real-time vision applications.
- **Build and optimize a CNC machine (4-axis foam cutter)** for precision cutting and submit to the International Symposium on LIFE MECHATRONICS
- **Design and develop a flight controller** for UAV applications.

PROFESSIONAL EXPERIENCE

2020-Present

- **Exchange student in France (ECAM LaSalle, 1 semester)**, studying industrial and mechanical engineering in an international environment.
- **Workshop at Kasetsart University, Thailand**, gaining hands-on engineering experience.
- **Teaching about Robotics** as a Volunteer at Passerelles Numériques (PN)
- **Industry visits** to Delta Electronics, MinebeaMitsumi, Phnom Penh Airport, and Coca-Cola, exploring emerging technologies.
- **Internship at Lamine Airline:** Simulation of Jet engine.
- **Participant in the Techno Innovation Challenge** and the 10th Engineering Innovation Challenge (Ministry of Education, Singapore).
- **Developed engineering projects**, including a 4-axis robotic arm, mini 3-wheel & 4-wheel cars, a car line-tracking system, an autonomous grass cutter, and water pump automation.
- **Mentor at Wander in Tech**, guiding aspiring engineers.
- **STEM team leader (3 years)** in high school
- **Study tour to universities in Taiwan and Vietnam** for research and innovation insights.

REFERENCES

Dr. SRANG SAROT

Head of Dynamics and Control Laboratory
/ ITC

Phone: (+855)85-689-858

Email: srangsarot@itc.edu.kh

SKILL

- SolidWorks
- Solid Edge
- AutoCAD
- Python
- Matlab
- C++
- KiCad
- PLC
- Problem-Solving
- Microsoft Office

LANGUAGE

- Khmer
- English

EDUCATION

Santhormok High School

(Secondary School + High School)

2015 - 2021 Phnom Penh

Robotics and Automation Engineering

Ecamlasalle

2024 - 2026 Phnom Penh

Industrial and Mechanical Engineering

Institute of Technology of Cambodia

2021 - 2026 Phnom Penh

LINKIN Account

Phone

(+855)-14-880-838

Email

primratanak@gmail.com

Address

St 138z, Terkleek 2, Toul Kork,
Phnom Penh

LINKIN Account:



Dr. YVON CHALM

K-box Solution / CEO

Phone: (+855)97-740-8080

Email: Yvon.chalm@k-boxsolution.com